

ORIGINAL RESEARCH

A complete in-cabin monitoring framework for autonomous vehicles in public transportation

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Abstract

Autonomous vehicles (AVs), driven by state-of-the-art deep learning and computer vision technologies, can revolutionize current mobility systems in modern transportation. Driverless AVs are slowly integrated into public transportation with significant advantages for the passengers and public transport operators. However, passenger safety and comfort are two of the main challenges that need to be addressed. This work presents a complete in-cabin monitoring framework with a suite of services, employing deep learning algorithms using a variety of onboard sensors at the edge. This proposed framework offers various innovative services aimed at enhancing security, monitoring passenger presence, accommodating diverse needs, and personalizing the passengers' travel experience, while also reducing the workload of human safety officers. Experimental results demonstrate the framework's effectiveness in identifying abnormal events with a high accuracy, employing multiple datasets and custom in-cabin scenarios. Additionally, the system effectively conducts automated passenger counting and facial identification, ensuring real-time responsiveness under diverse operational conditions. Overall, the novelty of this work lies in the framework's multimodal approach, integrating visual and audio analysis, to achieve robust performance across various scenarios, significantly contributing to the advancement of autonomous driving technologies.

1 | INTRODUCTION

Autonomous vehicles (AVs) are a significant technological advancement in the field of transportation, expected to deliver high levels of convenience, efficiency, and accessibility, while demonstrating progress in navigating complex road environments. One of the main reasons why AVs have emerged is their potential to reduce human errors and road accidents by respecting the traffic rules and regulations consistently. However, while the potential benefits of self-driving vehicles are vast, several concerns related to passenger safety and security have emerged that need to be addressed.

Passenger safety in AVs is a significant challenge because of the absence of the human driver or a human authority inside the shuttle. Unlike traditional human-driven cars, AVs require an automated way for monitoring and safeguarding passengers, being also our motivation for the development of various in-cabin monitoring approaches that are designed to continuously assess the state of the passengers and cabin conditions. The

main advantage of in-cabin monitoring systems is their ability to detect and respond to potential safety risks in real time, ensuring that passengers are safe and comfortable for the entirety of their journey.

In this context, the core motivation behind this work is to develop a comprehensive in-cabin monitoring framework for AVs that addresses crucial challenges related to passenger safety, security, and overall comfort. This paper presents a holistic approach to in-cabin monitoring, encompassing a range of services that contribute to the successful adoption of AVs by relying on a common foundation of hardware, sensors and state-of-the-art deep learning algorithms.

The primary objective of this proposed framework is to eliminate the need for a human safety officer within the AV by implementing advanced monitoring services to reduce the reliance on human intervention, leading to increased operational efficiency and reduced costs. Furthermore, the aforementioned hardware–software integration allows the provision of a seamless and comprehensive set of services, each playing a unique

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role in ensuring the success of AVs. Additionally, to address the security concern, an ‘augmented security and trust’ service is proposed that combines real-time surveillance, anomaly detection, and threat identification to create a secure in-cabin environment by utilizing state-of-the-art computer vision techniques to identify and respond to potential security breaches in real time. Another proposed service refers to the automated passenger counting that continuously tracks passenger movements and locations by employing deep learning algorithms and sensors to ensure that all passengers are accurately accounted during each trip, enhancing both safety protocols and operational reliability. This service is particularly beneficial for on-demand systems, where accurate real-time occupancy data are crucial for operational adjustments. Additional services regarding the safe transportation of children and elderly individuals have also been proposed. The ‘follow my kid/grandparents’ service was introduced, which leverages computer vision to provide real-time updates to concerned family members, assuring their loved ones of their well-being during their transport, contributing to a sense of security and trust in AVs. Ultimately, the proposed framework is specifically designed to be adaptable to both traditional public transportation systems and on-demand services, addressing the needs of different AV use cases.

The remainder of this paper is organized as follows. In Section 2, some related works from the most recent bibliography are presented regarding already proposed in-cabin monitoring systems to showcase the state that is so far reached. Section 3 describes all the services developed for this proposed multimodal in-cabin monitoring framework, while Section 4 describes the overall architecture of the proposed framework. Section 5 describes the methodology used and the design of each AI model followed by Section 6 which presents the experimental results of this study. Finally, Section 7 summarizes the conclusions of this work.

2 | RELATED WORK

AVs are becoming part of our lives, promising a shift in the way that transportation is perceived by enhancing mobility and efficiency on roadways. However, passenger safety is still one of the main challenges that needs to be addressed. For this purpose, in-cabin monitoring systems have been introduced, having the potential to increase the safety and comfort of the passengers inside self-driving cars and vehicles.

In this context, Mishra et al. [1] discussed the need for in-cabin monitoring in fully AVs to ensure the safety and security of both the passengers and the vehicle itself. More specifically, the authors proposed a single artificial intelligence (AI)-based camera system for monitoring and understanding in-cabin occupants and their behaviours, addressing various irregular situations that may occur inside the shuttle. In a later study, Mishra et al. [2] also focussed on the privacy challenges in in-cabin monitoring of AVs, addressing the need to balance the monitoring of the passengers’ behaviour and emotions for safety purposes, while simultaneously preserving their privacy. For this purpose, they proposed a novel framework

using a human–computer interaction (HCI) system and a generative adversarial network (GAN)-based approach for facial anonymization, allowing for the monitoring of the passengers’ emotions and intentions, while also providing facial anonymity. In their last iteration, Mishra et al. [3] proposed a robust in-cabin monitoring system that combines safety, security surveillance, and privacy concerns in AVs by incorporating AI- and camera-based systems for monitoring the cabin of higher-level AVs. Their study emphasized passenger privacy and compared various AI models for object and occupant recognition, while also introducing a comprehensive database to support research in seamless in-cabin monitoring AVs.

On the other hand, Abdulwahhab et al. [4] emphasized the use of deep learning for image analysis using techniques like image segmentation, classification and enhancement along with CNNs and GANs to achieve high accuracy and reliability, while Al et al. [5] proposed a model predictive and Stanley-based controller to achieve improved path-following and vehicle stability across diverse scenarios. Finally, Poon et al. [6] explored the application of You Only Look Once (YOLO) deep learning models for in-cabin monitoring using a fisheye-lens camera, focusing on detecting various in-cabin activities, including normal and distracted driving, as well as the presence of passengers and pets. The authors compared the performance of different YOLO-based models in terms of accuracy, speed and efficiency in real-time in-cabin monitoring, emphasizing the importance of high-performance object detection for enhancing driving safety in AVs.

It is evident that all of the aforementioned approaches share some main similarities, including their main scope that is in-cabin monitoring in AVs for enhancing security and passenger safety. In addition, they also leverage AI and deep learning models, such as YOLO and GANs, for object and occupancy recognition, behaviour analysis, and ensuring real-time monitoring efficiency, while simultaneously addressing privacy issues related to monitoring the passengers. However, despite their unique contributions to the field of in-cabin monitoring for AVs, all the studies described above focus only on passenger safety and anonymity, failing to propose additional in-vehicle services that could also be greatly useful for AVs, such as passenger counting and unaccompanied luggage monitoring. Furthermore, there is no other study to our knowledge that leverages different features, employing multimodal feature extraction and fusion by combining RGB, depth, and audio modalities.

3 | IN-VEHICLE SERVICES

The reasoning behind the proposed in-vehicle services is two-fold. Firstly, all of these services are considered a necessity for removing the safety officer from inside the shuttle, since safety and operational efficiency are two critical factors. In addition, all these services are based on the same technological foundation of using a combination of the same hardware with appropriate deep learning algorithms to provide all the expected results. In this section, an overview of the proposed

TABLE 1 Proposed operational scenarios related to each in-vehicle service.

In-vehicle service	Operational scenarios
Augmented security and trust	Passengers fighting
	Passengers falling down
	Bag snatch scenarios
	Vandalism scenarios
Automated passenger presence	Passenger counting
	Route optimization
	Passenger awareness
Follow my kid/grandparents	Night bus routes
	Kids and patients monitoring

four in-vehicle services is offered, namely augmented security and trust, automated passenger presence, and follow my kid/grandparents service.

Table 1 offers a comprehensive classification of each operational scenario presented with regards to the respective in-vehicle service.

3.1 | Security and trust

Typically, drivers were trained to handle in-cabin incidents of passengers' abnormal behaviour and petty crimes according to standard procedures adopted by the respective transport operator. However, with the emergence of AVs, in-vehicle surveillance is accomplished by using appropriate cameras and microphones along with state-of-the-art software so as to maximize both the efficiency and the actual level of in-cabin security. In this context, the 'security and trust' in-cabin service is proposed as a way to address the new reality that is formed in AVs as a result of the absence of the bus driver, as well as the increased threat from terrorism in European cities. It is responsible both for performing timely, accurate, robust, and automatic detection of various petty crimes or misdemeanours and assisting authorized end-users towards re-identifying any offenders. As a result, this solution supports safekeeping not only the passengers of the AV but also the vehicle itself by implementing video, depth and audio analytics modules for an embedded security framework that notifies both the passengers and the authorities in case of an abnormal event.

For the purpose of this study, four indicative operational scenarios have been explored for detecting passengers fighting and/or falling down, as well as identifying bag snatch and vandalism scenarios that may occur inside the AV shuttle, as follows:

3.1.1 | Passengers fighting

While traveling with the AV, a physical fight occurs between two passengers inside the shuttle. The advanced sensor array of the

vehicle, including cameras and microphones, continuously captures the action as it unfolds, while the system's deep learning algorithms quickly process these streams of raw data to determine the severity of the issue. Then, a notification is sent to the operator, whose actions upon assessing the severity of the incident from the video and audio feeds depend on the policies and procedures established by each transportation authority.

3.1.2 | Passengers falling down

During the journey, the AV suddenly stops. This unexpected stop causes passengers to lose their balance and fall on the floor. The AV's control system receives real-time information regarding this incident via the onboard sensors and the system's deep learning algorithms analyse the situation, recognizing the need for immediate action. As a result, the system sends an immediate alert to the operator, who may then evaluate the situation and take any necessary actions to resolve any underlying problems.

3.1.3 | Bag snatch scenarios

While a passenger is inside the AV, a petty crime takes place inside the vehicle in the form of an assault, with the passenger being attacked by another commuter who is attempting to steal their bag. This aggressive incident is captured by the in-cabin sensors (cameras and microphones) and fed into the system's deep learning algorithms for further analysis. Then, the operator receives the system's notification and acts accordingly, as the system is responsible only for alerting the authority in charge.

3.1.4 | Vandalism scenarios

An onboard passenger attempts to perform an illicit act of painting graffiti on the shuttle's windows. The incident is yet again captured by the onboard cameras, is fed into the system's deep learning algorithms for further analysis, and the system voices an audio warning through the AV's radio system or alerts the security personnel to stop the bus.

3.2 | Automated passenger presence

Traditionally, passenger counting is a process conducted manually via passenger surveys or with inspectors and drivers, something that is no longer feasible in AVs in the absence of the driver. On the other hand, newer, more simplistic approaches include implementing door sensors or log-in/out devices for real-time and continuous passenger monitoring, being however limited only to the vehicle's entry and exit points. The 'automated passenger presence service' is proposed as a way to address the in-cabin occupancy issue and raise awareness of the number of onboard passengers for better route scheduling, offering significant advantages over simpler alternatives. In this service, an automated system is introduced, capable of

detecting passenger presence in real time with high accuracy, which will ultimately help in counting onboard passengers and calculating vehicle occupancy. In addition, this service could also be beneficial to passengers, as they could be informed in advance of any seat availability in order to plan their trip accordingly. As a result, this approach offers a more comprehensive solution beyond basic occupancy detection through accurate passenger counting even in cases of mid-journey changes, such as unauthorized boarding or disembarking, and unaccompanied luggage and anomalous behaviours detection within the shuttle.

As in the previous service, three indicative operational scenarios have been proposed, respectively, including passenger counting, route optimization and passenger awareness:

3.2.1 | Passenger counting

The AV has a fixed capacity regarding the number of onboard passengers it should be carrying. For this purpose, the installed video cameras collect RGB and depth modalities from the in-cabin video feed, which are then fed into the system's deep learning algorithms for further processing. In case the total number of onboard passengers is reached, the shuttle stops receiving any additional users and sends appropriate notifications to excluded users regarding the reason why they could not board the AV.

3.2.2 | Route optimization

Even though the AV is at full capacity, there may still be people waiting at the bus stop to board the shuttle. Since the number of onboard passengers is continuously monitored, any new passengers can board the bus only in case others get off it. For this purpose, the AV modifies its route to save time and cost and will only stop in case a passenger needs to get off at their designated bus stop.

3.2.3 | Passenger awareness

Even though the AV has reached its terminus, there could still be onboard passengers. In this case, the AV counts the passengers present for one last time in order to notify them that they must disembark the shuttle.

3.3 | Follow my kid/grandparents

In addition to the aforementioned services, the 'follow my kid/grandparents' service is proposed to increase the autonomy of non-fully autonomous people, such as underage individuals or the elderly. The reasoning behind its design is twofold. Firstly, this service will reassure caretakers and close family members that their beloved family members are safe while moving around the city using public transportation. On the other hand, it can also have a positive impact on the passengers themselves by

helping increase their confidence in using public transportation knowing that their family is tracking them for the entirety of their trip by implementing a solution that continuously monitors their route.

In order to showcase the exact purpose of this service, three distinct operational scenarios have been proposed regarding the advantages of using this tracking solution on night routes and for monitoring both kids and patients, as follows:

3.3.1 | Night bus routes

Night bus routes can be unsettling for vulnerable individuals. For this purpose, the 'follow my kid/grandparents' service continuously collects RGB and depth modalities from the in-cabin video feed, feeds them into the system's deep learning algorithms for further processing and, once the passenger has been identified, the system's tracking begins and their caretakers can monitor their route.

3.3.2 | Kids and patients monitoring

Since traveling alone using public transportation for a kid can be quite unsettling, the 'follow my kid/grandparents' service offers the chance for its parents to continuously track the kid's route for the entirety of the bus's trip by using RGB and depth modalities from the in-cabin video feed. Kids are not the only passengers who may find difficulties commuting with public transportation. Patients are another group that may develop their own difficulties, especially those that require constant monitoring by their caretakers. For this purpose, the 'follow my kid/grandparents' service gives the caretaker the ability to continuously track their route once onboard by collecting RGB and depth modalities from the in-cabin video feed.

4 | FRAMEWORK ARCHITECTURE

The development of an accelerated multimodal AI framework for embedded devices proposed for in-cabin monitoring in AVs is driven by a number of factors. The diversity of data types, including images, audio and sensor readings, requires a multimodal approach. As presented in the following subsections, this integration allows for a comprehensive analysis of different datastreams, enhancing the devices' ability to interpret complex scenarios and make informed decisions.

4.1 | Hardware platform

The system is optimized for the NVIDIA Jetson AGX Xavier platform, but it is adaptable to similar hardware configurations, requiring at least an 8-core Cortex-A72 CPU, GPU with CUDA support, 16 GB RAM and 32 GB disk space. Software prerequisites include an Ubuntu-based ARM64 OS, Python and OpenCV. The installation process involves system

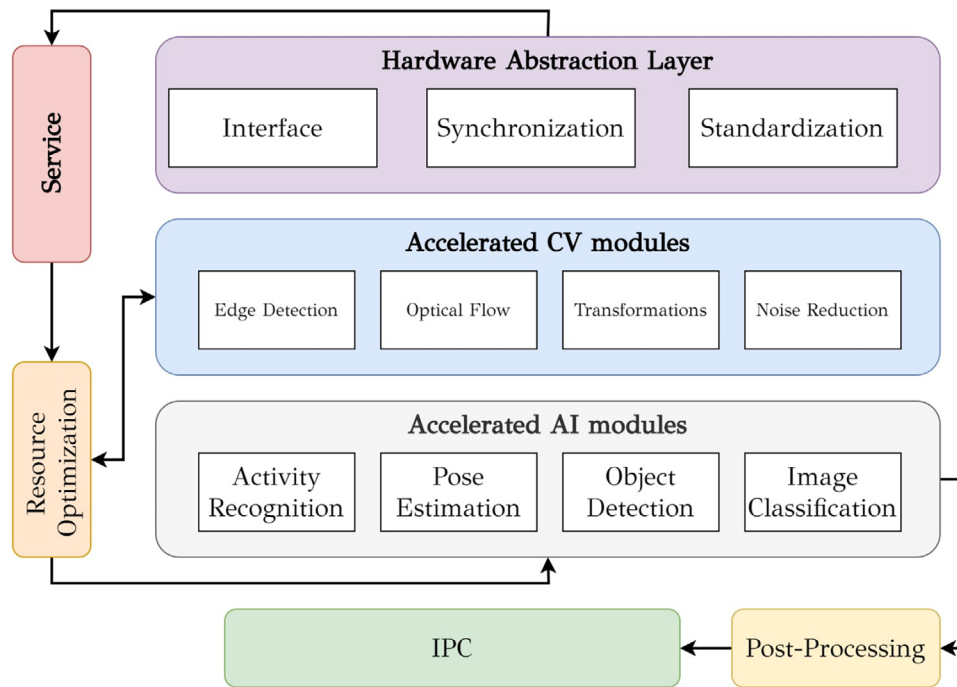


FIGURE 1 Framework-based service architecture.

package installations, building the `v4l2loopback` device for creating virtual camera devices and installing necessary Python packages. The structure is comprised of various directories for assets, common code, service modules, shared modules, utilities, a configuration file and a launcher script. The configuration settings are adjustable in the `config.py` file, which includes constants such as shared modules, service modules, vehicle ID, camera sources and parameters for different services, including passenger counting, object detection and face verification.

The system is launched by spawning service modules as separate processes and allowing inter-process communication through the `av_ipc` module. The dashboard, accessible locally or over a network, displays video feeds, audio spectrograms, status of different modalities, event history, environmental metrics, passenger counts and other crucial data related to security and monitoring in AVs. The framework is designed to be robust, adaptable and user-friendly, catering to the complex needs of AV control and monitoring systems.

4.2 | Interfaces

The framework, as depicted in Figure 1, integrates an advanced hardware abstraction layer (HAL), designed to interface seamlessly with a diverse array of sensors. This intermediate layer is responsible for standardizing communications and data exchange between the processing units and various external hardware devices. This level of abstraction not only simplifies integration but also enhances the system's adaptability and scalability. For visual data acquisition, the HAL supports a range of USB camera types, while being capable of interfacing with

IP cameras using multiple protocols such as RTSP, HTTP and ONVIF, ensuring compatibility with a wide array of network-based camera systems. This flexibility allows the framework to be deployed in environments with pre-existing surveillance infrastructure, or in scenarios requiring remote or distributed monitoring setups. In addition to its wide sensor support, the framework's HAL incorporates an advanced data synchronization mechanism, crucial for coordinating inputs from multiple sensors. This synchronization ensures that datastreams from diverse sources, such as IP cameras, USB microphones and various other sensors, are aligned in time, providing a coherent and unified view of the environment. This temporal alignment is vital in scenarios where the timing of events captured by different sensors is critical. The synchronization module intelligently manages time-stamps, handles latency variations and aligns data sequences, thereby maintaining the integrity and relevance of the information being processed. This capability not only enhances the accuracy and reliability of the system's analyses and responses but also enables more sophisticated functionalities including event detection and multimodal data fusion.

5 | METHODOLOGY

Deep learning models are efficient in handling complex tasks, especially in computer vision. The abundance of data from camera sensors renders deep learning approaches suitable for large and diverse datasets. In this section, the proposed approaches for analysing multi-sensory data are presented, captured by RGB, depth cameras and microphone arrays. This section is specifically focused on analysing patterns for detecting

abnormal video and audio events, counting passengers, identifying facial characteristics and performing gesture recognition.

5.1 | Video-based abnormal event detection

In this section, we present multiple approaches for video abnormal event detection, such as bag snatch, vandalism, as well as passenger fighting and falling down scenarios. The first three single-modality (RGB) approaches are based on pose classification, spatiotemporal autoencoders and hybrid long short-term memory (LSTM) classification models, as presented in detail in our previous work [7]. We perform pose classification in a four-stage process, starting with extracting 15 key body points for each person in the frame and tracking them across the sequences using the Euclidean-based distance. The third stage converts these keypoints as trajectories, while the last stage includes inference by the classifier. This method uses a combination of pose estimation frameworks and LSTM neural networks for feature extraction and classification.

On the other hand, the spatiotemporal autoencoder approach, based on the principle that recent frames will significantly differ from older ones in abnormal events, uses a spatial feature extractor and a temporal encoder–decoder to learn the temporal patterns of input frames. The model is trained with normal scenes to minimize reconstruction error. A low reconstruction error in normal videos contrasts with high error in abnormal scenes, enabling abnormal event detection through thresholding on the error produced. Finally, the hybrid LSTM classification model uses a semi-supervised learning model that combines a high recaller (autoencoder) and a false positive reduction model (hybrid model), specifically designed to detect almost all actual anomalies (high recall) along with some false positives (low precision). The training involves three stages: training the autoencoder with unsupervised data, transferring encoder weights to the hybrid model, and supervised training using only the LSTM classifier.

However, in latest experiments, we implemented an additional two-stream action classification model. This approach, is based on the SlowFast [8] algorithm, which employs parallel streams with different frame rates: A slow pathway for slow motions with high spatial detail and a fast pathway for quick-changing motions with less spatial detail, as presented in Figure 2. This algorithm processes rectangle RGB images and uses the Adam optimizer with a categorical cross-entropy loss function. The model's dual-pathway approach effectively captures both slow and fast actions, making it suitable for action recognition in various temporal contexts.

The depicted architecture (Figure 2) utilizes a dual-pathway approach to process video frames for classification tasks. The slow pathway operates on the video frames at their native temporal resolution T , applying 3D convolutions (denoted by $\text{Conv}(3 \times 3 \times 3)$) to capture spatial features in high fidelity over time. This pathway is designed to extract detailed spatial-temporal information due to its access to the full temporal spectrum of the frames. In contrast, the fast pathway processes a temporally subsampled version of the frames, reducing the

resolution to αT , while simultaneously increasing the channel capacity to βC . This is achieved by applying a subsampling factor α , which allows the pathway to emphasize rapid temporal variations by focusing on the evolution of the content between subsampled frames. Both pathways are designed to converge into specialized attention blocks that recalibrate the feature maps. The spatio-temporal (ST) attention block integrates both spatial and temporal dimensions, employing a convolution operation $\text{Conv}(1 \times 1 \times 2)$, to emphasize salient features and enhance the representational capacity of the network. Similarly, the temporal attention block is tailored to refine the temporal features, utilizing a convolution operation $\text{Conv}(1 \times 3 \times 2)$ that focuses on temporal dynamics for capturing motion and changes across the video sequence. Upon fusing the outputs from both attention mechanisms, the combined feature maps are processed through additional convolution operations before classification. The final classification is performed by a classifier, which outputs the probability distribution across the target classes.

In this framework the two-stream classification method has been used for abnormal event detection due to its suitability for the specific demands and characteristics of the AV environment. More specifically, the dual pathway implementation is a more refined approach to capturing the complex dynamics of in-cabin activities by employing dual pathways that operate at different frame rates, adeptly capturing both rapid and subtle movements within the cabin. Such a dual-pathway system is crucial for AV environments, where both rapid movement (such as a fighting event) and slower, more gradual motion (such as painting a graffiti), are equally important for efficient recognition.

5.2 | Automated passenger presence

Inspired by RAPiD [9], the automated passenger presence and unaccompanied luggage detection network consists of three stages, namely the backbone network, the feature pyramid network (FPN), and the bounding box regression network, as depicted in Figure 3. Similar to one-stage detectors, the proposed overhead imaging model contains a Backbone network, an FPN and a bounding-box regression network as a detection head, with each component specifically designed to process the input image and extract features at multiple scales, resulting in the prediction of bounding boxes and class categories for objects within the image. The backbone of the network is responsible for the initial feature extraction and is typically a pre-trained CNN such as ResNet, VGG, or a similar architecture. More specifically, given an input image I , the network produces a set of multi-dimensional feature maps $\{P_k\}_{k=1}^3$ at different scales, denoted as P_1 , P_2 , and P_3 for high, medium, and low resolutions, respectively, as follows:

$$P_k = \text{Backbone}(I), \quad k \in \{1, 2, 3\} \quad (1)$$

The FPN enhances the backbone's feature maps by integrating high-level semantic information from deep layers with spatial information from earlier layers based on Equation (2).

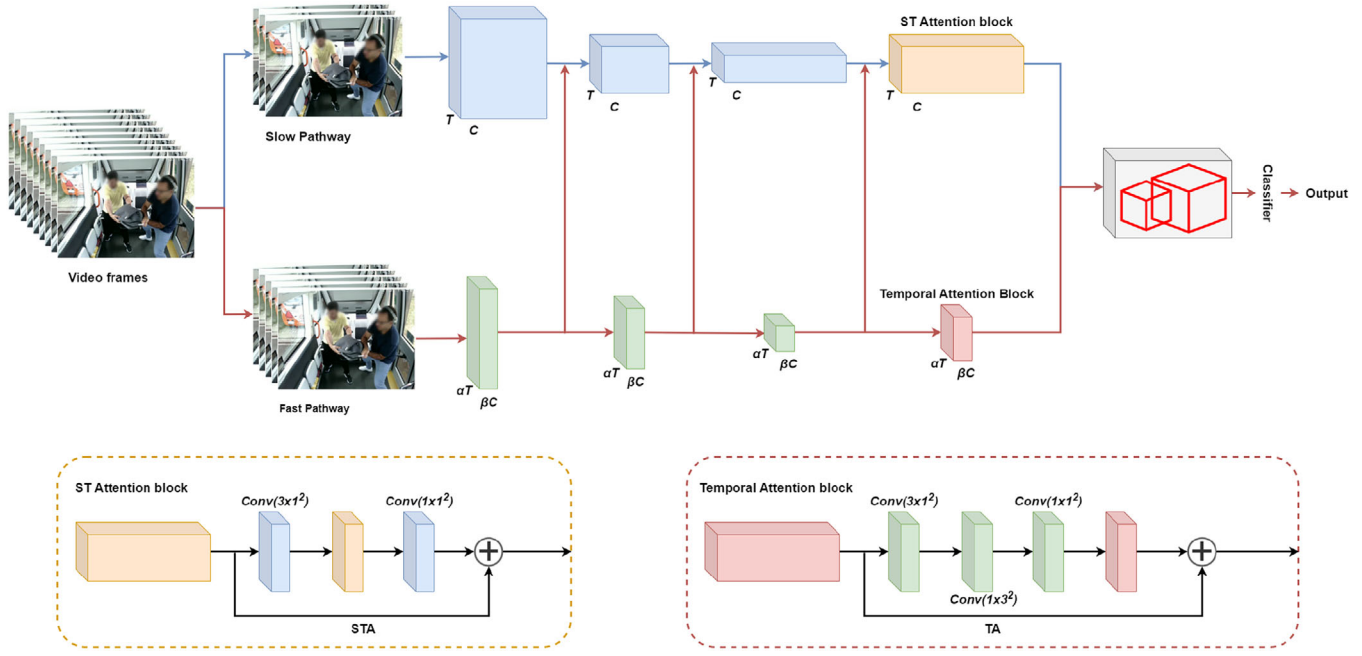


FIGURE 2 Proposed dual-stream network for abnormal event detection in autonomous vehicles.

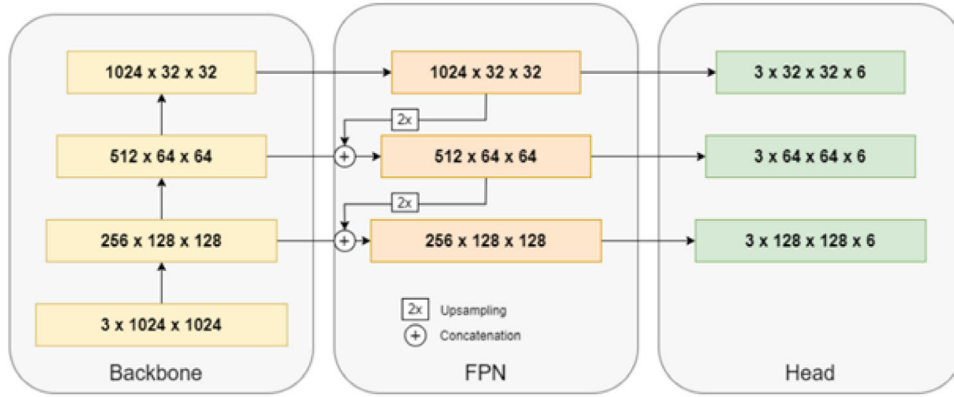


FIGURE 3 An illustration of multiple convolutional layers and multi-dimensional matrices such as the feature maps with 1024 × 1024 input resolution.

This is achieved by upsampling spatially coarser, but semantically stronger, feature maps from higher pyramid levels and merging them with lower-level maps through element-wise addition.

$$\{P_k^{\text{fpn}}\} = \text{FPN}(\{P_k\}), \quad k \in \{1, 2, 3\} \quad (2)$$

Post feature enhancement by the FPN, the detection head predicts the transformed bounding boxes and class scores by applying a separate CNN to each FPN feature vector to generate a transformed bounding box predictions $\tilde{T}_k$ and an objectness score, which are relatively parameterized to anchor pre-defined boxes at different scales and aspect ratios, as follows:

$$\tilde{t}_{x,k} = s_k \cdot (i + \text{Sigmoid}(f_{x,k})) \quad (3)$$

$$\tilde{t}_{y,k} = s_k \cdot (j + \text{Sigmoid}(f_{y,k})) \quad (4)$$

$$\tilde{t}_{w,k} = w_k^{\text{anchor}} \cdot \exp(f_{w,k}) \quad (5)$$

$$\tilde{t}_{h,k} = h_k^{\text{anchor}} \cdot \exp(f_{h,k}) \quad (6)$$

$$\tilde{o}_k = \text{Sigmoid}(f_{o,k}), \quad (7)$$

where i and j represent the centre of the anchor box, s_k is the stride of the feature map at scale k , and $f_{x,k}, f_{y,k}, f_{w,k}, f_{h,k}, f_{o,k}$ are the outputs of the CNN applied to the feature vector from the FPN. The variables w_k^{anchor} and h_k^{anchor} represent the width and height of the k -th anchor box, and Sigmoid is the logistic function applied to constrain the outputs to a range between 0 and 1. The objectness score $\tilde{o}_k$ denotes the probability that an object is present within the predicted bounding box. The

training loss function of the network combines the regression loss for the bounding box coordinates and a classification loss for the objectness score.

5.2.1 | Rotation-aware bounding box regression

The rotation-aware bounding box regression is a critical component of the proposed method, enabling the detection of people in overhead fisheye images with a high degree of accuracy in both position and orientation by incorporating a novel angle prediction mechanism that accounts for the unique properties of angles as cyclic quantities. It involves the prediction of a bounding box's orientation, along with its centre and size, which are normalized relative to the feature map dimensions as follows:

$$\mathcal{L}_{\text{reg}} = \sum_i^N \left(\lambda_{\text{coord}} \text{Scale}_{L1}(b_i, \hat{b}_i) + \lambda_{\text{angle}} \text{Periodic}_{L1}(\theta_i, \hat{\theta}_i) \right) \quad (8)$$

Here, $\mathcal{L}_{\text{reg}}$ represents the regression loss, b_i the predicted bounding box, $\hat{b}_i$ the ground truth box, θ_i the predicted orientation, and $\hat{\theta}_i$ the ground truth orientation. The terms Scale_{L1} and Periodic_{L1} are the scale $L1$ loss and periodic $L1$ loss, respectively, with λ_{coord} and λ_{angle} as the balancing weights. This allows the network to effectively learn the orientation of objects, taking into consideration the periodic nature of the angle, thus providing a more robust and accurate object detection in fisheye images.

5.3 | Audio-based abnormal event detection

CNNs are generally known to perform well in image classification and have achieved similar results in the field of audio-based event detection, hence were chosen in this proposed audio abnormal event detection framework. In this section, the CNN architectures used, the optimizer that was selected and the activation function between the convolutions and max-pooling operations are described. For audio classification a DenseNet-121 CNN architecture is used as seen in Figure 4.

DenseNets [10] were introduced in order to solve the problem of the vanishing gradient in neural networks, which occurs when the CNNs become big that the path for information from the input layer to the output layer significantly increases that the gradient vanishes during back-propagation. In DenseNets, every layer is directly connected with each other, ensuring maximum information kept throughout the architecture. Additionally, such connections allow the network to require fewer parameters to train than vanilla CNNs, since there is no need to learn redundant feature maps. The DenseNet-121 was used for classification, as the most basic DenseNet yet powerful architecture and each dense layer consists of two convolutional operations: 1×1 convolutions for feature extraction; and 3×3

convolutions for bringing down the feature depth/channel output. Finally, the DenseNet-121 consists of six such dense layers in a dense block, which resulted in approximately seven million parameters, compared to the 44 million parameters of a vanilla CNN architecture. In our experiments, we have used the Adam optimization method with an initial learning rate of 0.001. Adam is often the preferred optimization method in machine learning, due to its computational efficiency, little memory occupation and implementation ease. Adam is a momentum-based method that only relies on the first derivative of the cost function. Required inputs are exponential decay rates β_1 and β_2 , cost function $c(\theta)$ and initial weights θ , in addition to the learning rate η and eventually a regularization λ . Iteratively, the gradient of the cost function is calculated and this is used to calculate the first and second moment estimates. The activation function is used to activate the outputs, which often need to take some certain properties. For example, when doing classification, the outputs are probabilities and therefore take values between zero and one. A non-linear activation function is often preferred to reinforce the non-linearity of the neural network. Traditionally, the logistic function and the $\tanh$ function have been used as activation functions, since they were believed to work in the same way as the human brain. However, in 2012 Alex Krizhevsky et al. introduced AlexNet [11], taking image recognition to a new dimension. They used a ReLU function, which is zero for negative values. The ReLU function is a modification of the pure linear function for positive values. This makes the function able to recognize the non-linearity in the model, providing it a 'clean' derivative given by the step function.

5.4 | Follow my kid/grandparents

For the purposes of the 'follow my kid/grandparents' in-vehicle service, a facial identification framework has been developed. However, as current face verification models use fully connected layers, spatial information is lost along with the ability to understand the convolution features in a human sense. To address this obstacle, the plug-in xCos module is integrated that is comprised of three distinct stages, namely the input stage, the backbone stage, and the Lcos calculation (xCos) stage, as presented in Figure 5. In the first stage, the two input images are preprocessed and passed into the feature extractor, where input A corresponds to the database image while input B is the image cropped from the video stream. For the backbone stage, a CNN feature extractor has been implemented in the same way as in ArcFace [12]. However, in order to employ the xCos module, the last fully connected layer and the previous flatten layer are replaced with a 1×1 convolutional layer. Finally, in the last stage, the patch-wise cosine similarity is multiplied by the attention maps and then summed to calculate the Lcos.

Given two facial images A and B , let $F_A = \{\vec{f}_{A1}, \vec{f}_{A2}, \dots, \vec{f}_{An}\}$ and $F_B = \{\vec{f}_{B1}, \vec{f}_{B2}, \dots, \vec{f}_{Bn}\}$ represent the sets of grid features extracted from each image, where $\vec{f}_{Ai}$ and $\vec{f}_{Bj}$ are the feature vectors corresponding to specific facial regions or patches within images A and B , respectively. The modified xCos metric,

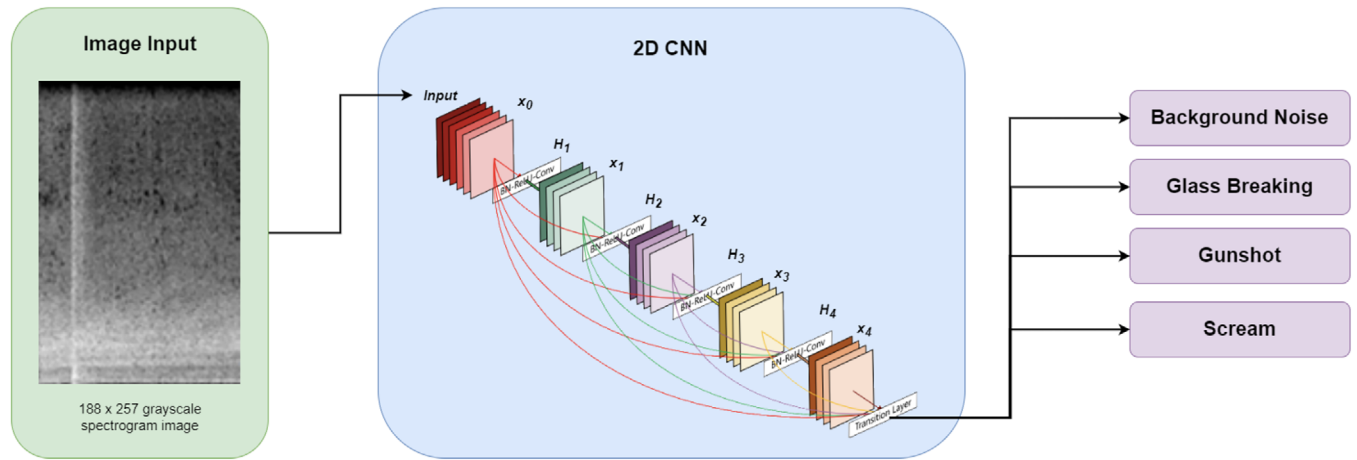


FIGURE 4 Pipeline of the 2D CNN used for the audio-based abnormal event detection.

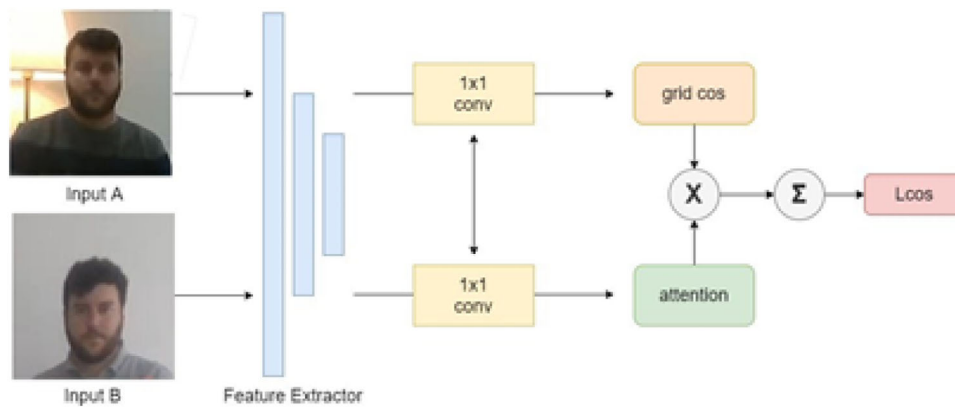


FIGURE 5 Facial identification network pipeline: The two input images are preprocessed and passed into the backbone CNN for feature extraction along with the plug-in xCos module.

incorporating an attention mechanism, is defined as:

$$xCos(F_A, F_B, W) = \sum_{i,j} w_{ij} \cdot \cos(\vec{f}_{Ai}, \vec{f}_{Bj})$$

Here, W is the attention matrix where w_{ij} denotes the weight assigned to the comparison between feature vectors $\vec{f}_{Ai}$ and $\vec{f}_{Bj}$, and $\cos$ represents the cosine similarity between these vectors, highlighting their angular proximity.

The attention weights W are dynamically learned, focusing on enhancing the contribution of more invariant and distinctive facial features to the overall similarity score, while minimizing the impact of variable aspects such as hairstyles or expressions. This process allows for a deep learning model to not only compare faces with high precision but also to provide insights into which features are most relevant for the comparison, thereby improving both the interpretability and reliability of the face verification task. To optimize and learn the attention weights in a practical setting, CNNs are employed for feature extraction, coupled with a dedicated attention module for computing W . The entire system is trained end-to-end, utilizing the xCos metric as a component of the loss function to refine

feature extraction and attention allocation, thus directly contributing to the model's ability to discern and emphasize the most informative features for face verification.

6 | EXPERIMENTAL RESULTS

In this section, the experimental results from the proposed multimodal in-cabin monitoring framework for public transportation are presented, focussing on evaluating the framework's effectiveness in accurately detecting and classifying a wide range of in-cabin activities, both normal and abnormal, under various operational conditions. The experiments were meticulously structured to assess the framework's robustness, accuracy, and real-time responsiveness, key factors crucial for ensuring passenger safety and comfort in autonomous driving environments. Utilizing a diverse dataset that includes various in-cabin scenarios, we have conducted extensive tests to measure the performance of our multimodal approach, which integrates advanced sensor technologies and sophisticated machine learning algorithms. This section will present the experimental results regarding both the visual- and the audio-based abnormal event



FIGURE 6 Evaluation of the proposed algorithm on several operational scenarios. Each column contains two images of fighting, bag-snatching, vandalism and falling-down events, respectively.

detection analysis, as well as the automated passenger counting and the facial identification analysis.

6.1 | Datasets

Regarding the abnormal event detection service, the proposed visual model was tested both on the NTU RGB+D dataset [13] and on the data captured by CERTH inside the AV's shuttle. Some indicative detection results are presented in Figure 6, where each column contains two images of fighting, bagsnatch, vandalism and falldown events, respectively. The datasets conform to ethical data collection practices, with each passenger entry accompanied by a documented consent form authorizing the use of their data. Moreover, the faces of the passengers were blurred to preserve the privacy of the people involved.

Regarding the automated passenger counting service, the proposed model was tested on using both the BOSS dataset and a custom one captured by HOLO and TPG, as presented in Figure 7. Although the scenarios illustrated showcase low vehicle occupancy, the system has been thoroughly tested under diverse conditions, including scenarios with full vehicle occupancy. The camera system is the D-Link DCS4625 panoramic fisheye camera with $f/2.0$ wide-angle panoramic lens and 5 MP (2560×1920 @30 fps) high-resolution video. The implemented algorithm monitors the shuttle in real time and records the position of each person in frames, as depicted in Figure 8. In addition, a custom mask is applied on large reflective surfaces to mitigate reflections in cases where passenger figures might appear to the windows of the shuttle as reflections.

6.2 | Main results

6.2.1 | Security and trust (video module)

For the abnormal event detection model, Table 2 represents a modified classification report with a total accuracy of 95.2%.

TABLE 2 Video analysis experiment: Classification report.

Class	Precision	Recall	F1 score
Fall-down	0.93	0.90	0.92
Fighting	0.82	0.86	0.84
Bag-snatching	0.80	0.85	0.82
Normal	0.96	0.98	0.97
Vandalism	0.94	0.92	0.93
Total accuracy			0.95

Within this report, the classification performance of different classes is evaluated. Notably, the 'fighting' and 'bagsnatch' classes exhibit similarities in the events they represent, which has led to confusion within the network's predictions. In the context of this classification problem, 'fighting' and 'bagsnatch' are two classes that share certain commonalities in visual characteristics or patterns, such as sudden and aggressive movements. Consequently, the network finds it challenging to distinguish between these two classes, resulting in similar precision, recall, and F1 score values for both. In this scenario, the precision, recall, and F1 score values for 'fighting' and 'bagsnatch' reflect this inherent confusion and are slightly lower compared to other classes. This classification report underscores the need for further refinement and feature engineering to enhance the network's ability to discriminate between these similar event classes. Achieving a total accuracy of 95% requires addressing the specific challenges posed by such visually overlapping classes, ultimately improving the network's performance in real-world scenarios.

6.2.2 | Security and trust (audio module)

Regarding the audio-based abnormal event detection analysis, a batch size of 16 images was used while the initial epochs were set to 200. However, it was observed that for the



FIGURE 7 Compilation of various frames for abnormal events from the three custom datasets (CERTH, TPG and HOLO+TPG).

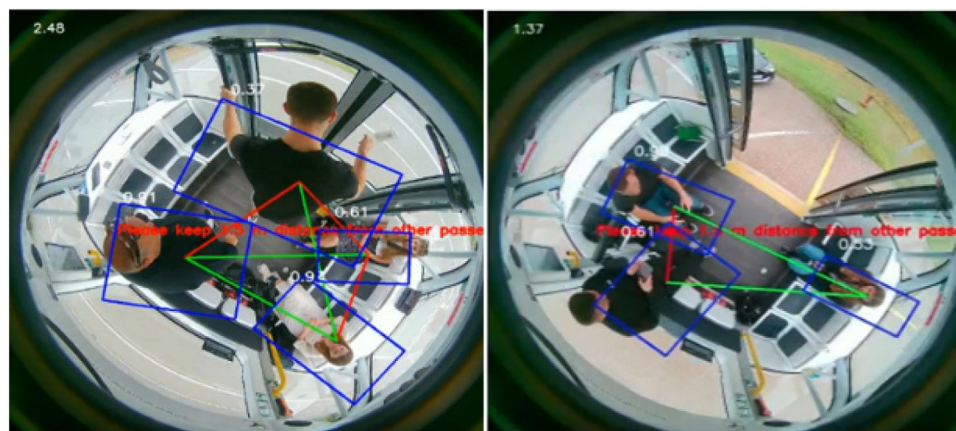


FIGURE 8 Results of the automated passenger counting service. Blue bounding boxes illustrate the detected passengers while the number above showcases the detection confidence on a [0–1] scale.

DenseNet-121 only eight epochs were sufficient to achieve the optimal performance, hence, an early stopping function was implemented to check the validation $F1$ macro averaged score for an improvement in five consecutive epochs. If no improvement was detected the network stopped the training in order to

avoid overfitting. The experimental audio results showcased that the DenseNet-121 achieved a training $F1$ -score of 95.92% and a validation $F1$ -score of 88.74% for the case of 0 dB SNR, while for 30 dB, the DenseNet-121 achieved a training $F1$ -score of 96.84% and a validation $F1$ -score of 91.82%. As expected, the

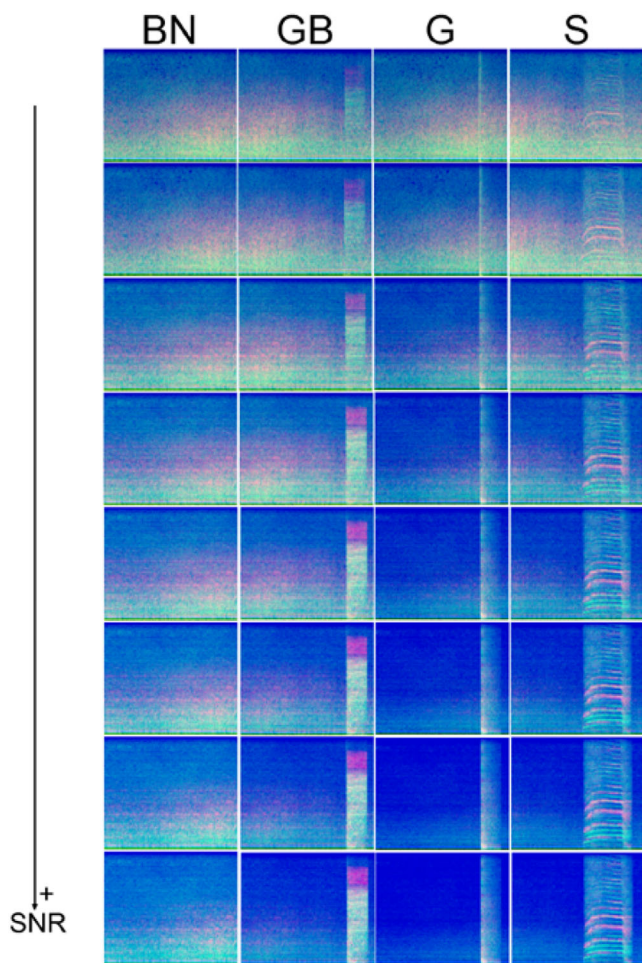


FIGURE 9 Multichannel spectrograms obtained with the stacked method: The representations of the four classes of the extended dataset are shown, with the SNR value increasing at the direction of the arrow (left side) from -5 dB to 30 dB with a step of 5 dB.

network is able to classify the target classes with higher accuracy in the case of 30 dB SNR and we notice that the training loss starts at smaller values, compared to the case of 0 dB SNR. A representative sample of the variation with respect to each class and SNR in the extended dataset is depicted in Figure 9. It is evident that, as the SNR increases, the features become clearer and easier to distinguish from the background, as was expected. The two common representations mentioned in the literature are spectrograms and gammatonegrams. The former is the traditional time-frequency visualisation, but it actually has some important differences from how sound is analysed by the ear; most significantly, the ear's frequency sub-bands get wider for higher frequencies, whereas the spectrogram has a constant bandwidth across all frequency channels. A gammatone spectrogram or gammatonegram is a time-frequency magnitude array based on an FFT-based approximation to gammatone sub-band filters, for which the bandwidth increases with increasing central frequency. The upper part of the table compares the results achieved by considering the classification of positive SNR sound events only are shown. In the lower part of the table, the results

achieved by including sound events with negative and null SNR to the above are exhibited. The average RRs for the three classes of interest (event-based) were 92.5% and 90.9% for the original and the extended dataset, respectively. The latter compares well with the reported value of 90.7% in [14].

6.2.3 | Passenger counting

Table 3 details a performance comparison of object detection methods across three datasets (HABBOF, CEPDOF, and AVOF). The evaluation utilizes the powerful NVIDIA RTX 4090 GPU. Performance analysis includes average precision at 50% IoU (AP_{50}), precision (P), recall (R), $F1$ -score (F), and inference speed in frames per second (FPS). Input resolutions are scaled and denoted in parentheses. All methods used a 0.3 confidence threshold. Our proposed method excels in the CEPDOF and AVOF datasets for accuracy. It maintains a competitive AP_{50} on HABBOF (97.9%) and a strong frame rate (29.1 FPS), proving its balance of accuracy and real-time suitability. RAPiD (1024) achieves the highest AP_{50} on HABBOF (98.1%), potentially aided by the dataset's emphasis on upright human poses, but exhibits a lower FPS (27.7). RAPiD (608) offers the fastest FPS (52.5), ideal for applications prioritizing real-time speed over some accuracy compromise. Our method has slightly reduced performance on HABBOF, suggesting potential avenues for improvement in handling radially oriented humans in fisheye images. Overall, our method demonstrates robust accuracy across diverse datasets, all while maintaining a high processing speed. This makes it a compelling choice for various object detection scenarios where both accuracy and real-time performance are critical.

6.2.4 | Follow my kid

The efficiency of face recognition algorithms in the presence of occlusions such as face masks was evaluated by testing on the MS1M-ArcFace and LFW datasets, along with their masked counterparts MS1M-ArcFace+M and LFW+M. Table 4 summarizes the performance metrics. The original unmasked datasets yielded high accuracy, with ArcFace achieving 99.83% on LFW. However, the introduction of the xCos module resulted in a slight decrease to 99.35% . When masks were applied into the datasets, the challenges in feature extraction became evident with a significant drop in accuracy. ArcFace, without any adaptations for masks, showed a decline in performance, having an accuracy of 68.33% on the MS1M-ArcFace+M dataset. Nevertheless, the integration of the xCos module with mask adaptation (ArcFace-xCos+M) mitigated this performance loss, improving the accuracy by approximately 6.2% reaching 74.52% . This enhancement signifies our method's efficiency in handling occlusions by employing patch-wise cosine similarity and attention mechanisms to extract facial features even when partially obstructed.

TABLE 3 Comparative analysis of different approaches utilizing the HABBOF, CEPDOF, and AVOF datasets, conducted on an RTX 4090 platform. The numbers within brackets represent the resolution of input, adjusted through multiplication by a power of two.

Method	FPS	HABBOF				CEPDOF				AVOF			
		AP ₅₀	P	R	F	AP ₅₀	P	R	F	AP ₅₀	P	R	F
(608) [15]	51.2	87.3	0.97	0.82	0.89	61.0	0.96	0.52	0.63	62.5	0.94	0.81	0.87
AA (1,024) [16]	3.2	87.7	0.92	0.86	0.89	73.9	0.89	0.63	0.68	75.0	0.95	0.82	0.88
AB (1,024) [16]	1.7	93.7	0.88	0.93	0.90	76.9	0.88	0.69	0.74	77.5	0.95	0.82	0.88
RAPiD (608) [9]	52.5	97.3	0.98	0.93	0.95	82.4	0.97	0.82	0.89	85.0	0.96	0.85	0.90
RAPiD (1,024) [9]	27.7	98.1	0.97	0.96	0.96	85.8	0.97	0.82	0.89	87.0	0.97	0.85	0.91
Proposed (1,024)	29.1	97.9	0.96	0.93	0.95	86.1	0.97	0.96	0.97	92.3	0.96	0.94	0.94

TABLE 4 Accuracy comparison between methods and datasets with and without masks.

Method	Training dataset	Testing dataset	Masks	Accuracy
ArcFace	MS1M-ArcFace	LFW	No	99.83%
ArcFace-xCos	MS1M-ArcFace	LFW	No	99.35%
ArcFace+M	MS1M-ArcFace+M	LFW+M	Yes	68.33%
ArcFace-xCos+M	MS1M-ArcFace+M	LFW+M	Yes	74.52%

TABLE 5 Number of samples validated and accuracy and *F1*-scores for each abnormal event scenario.

Event	Samples	Accuracy	<i>F1</i> -score
Bag-snatching	5	0.9	0.9091
Fighting	4	0.8889	0.8889
Fall-down	6	0.8333	0.8571
Vandalism	5	0.9167	0.9091

6.3 | Evaluation on real-world scenarios

The system was evaluated in automated minibuses in Copenhagen and Geneva. The solution was installed on NAVYA autonomous vehicles, featuring a NVIDIA Jetson AGX Xavier platform and a D-Link DCS-4625 fisheye camera. The camera was connected directly to the Jetson system via Ethernet through RTSP protocol. Both components are powered through the vehicle's batteries and Tensor-RT conversion was performed to maximize the algorithm's efficiency, reducing the power consumption to approximately 10 W.

Regarding the security and trust module, 20 video segments of 3 s with abnormal events and 20 video segments with normal data were inferenced in real time. Specifically, in Table 5, the number of samples validated for each class are presented, along with the accuracy and the *F1*-score metrics, calculated based on the inference result. As seen from the table, the results indicated that the algorithms were able to correctly identify

TABLE 6 Number of samples validated and accuracy and *F1*-scores for each passenger facial identification class.

Class	Samples	Accuracy	<i>F1</i> -score
Enrolled male	3	0.8571	0.8571
Enrolled female	2	0.8	0.8
Unknown male	5	0.9167	0.9091
Unknown female	4	0.875	0.8889

most of the performed scenarios with 89% accuracy, with the appropriate notifications being also captured in the operator's dashboard.

A simple algorithm was developed for the validation of passenger counting in order to sum the total number of passengers, concentrating only on increases in the passenger number. As depicted in Figure 10a, before the passenger counting service integration, the on-site shuttle operator was manually counting passengers using a dedicated application, which served as a basis for validating the accuracy of an automated passenger counting system by comparing it with the data transmitted from the operator's mobile device.

A real-world evaluation of the counting accuracy was conducted by analysing all the data collected during a continuous two-day operational phase, focussed on instances where the automated system detected more than one passenger (excluding the driver) in the shuttle. These instances were then compared, revealing patterns that indicate a similar passenger count, as depicted in Figure 10b. However, since automated counting data showcase constant fluctuations, they can also lead to irregularities in the overall onboard passenger number.

Finally, for the facial identification service results, 25 video segments of 3 s with facial data of five passengers were enrolled into the service and 25 segments with unknown passengers were inferenced in real time, as presented in Table 6, along with the accuracy and the *F1*-score metrics, calculated based on the inference result. As seen from the table, the results indicate that the algorithms were able to correctly identify the enrolled passengers with an average 86% accuracy.

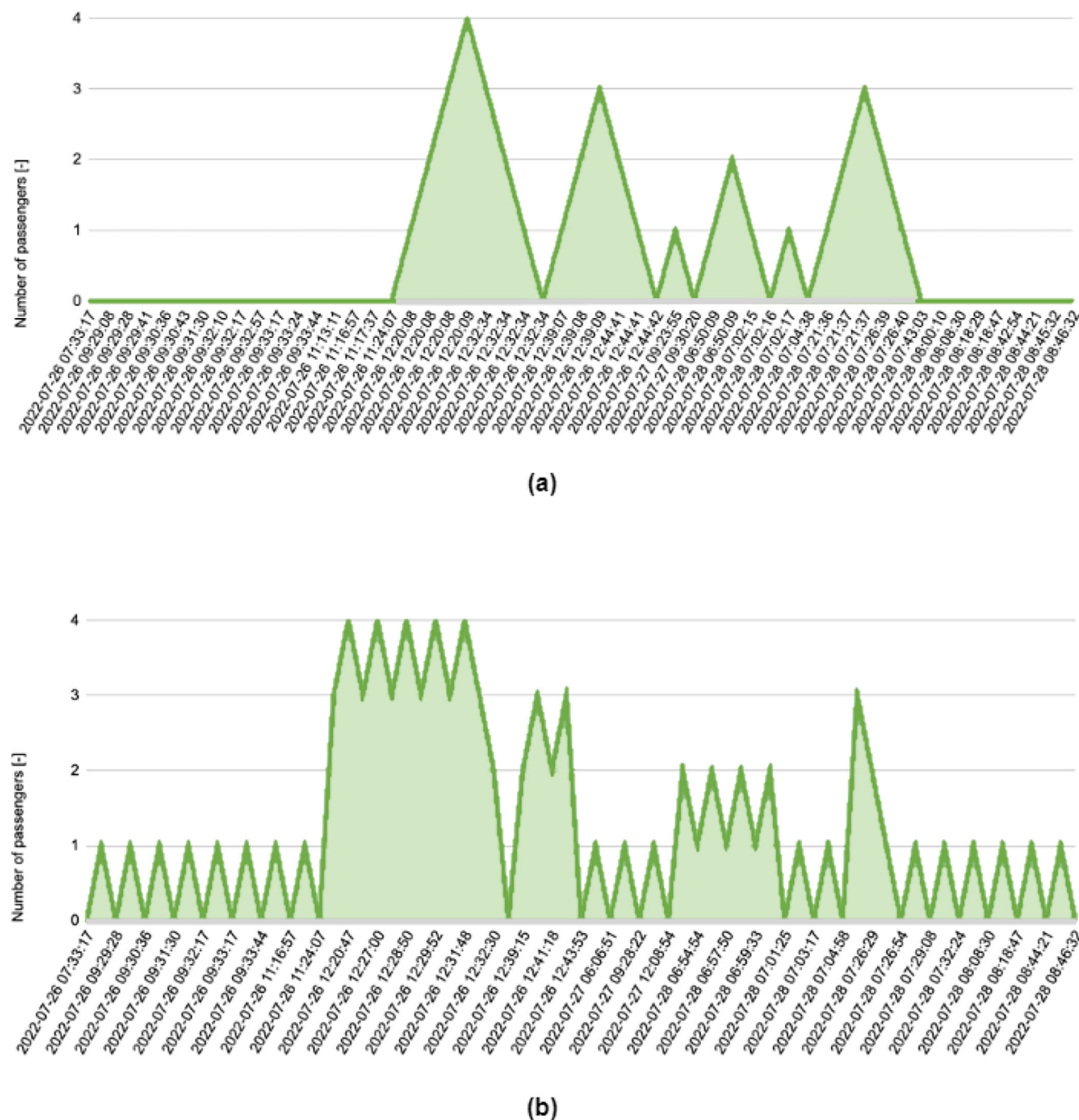


FIGURE 10 The number of passengers recorded over a period of time, received from two sources. (a) Manual data collected by the driver and (b) an automated data collection system.

Finally, for the facial identification service, the proposed model was designed as a flexible and end-to-end service accessible by an Android mobile application that involves three entities, namely the shuttle environment, the cloud platform, and the mobile application. The process begins with the enrolment of a new passenger by the client via the mobile application, using a single image of their face, which will then be processed and stored temporarily in a database on the cloud platform. By using

this as the reference image, the network calculates the similarity for any new instance presented to it from the shuttle. The network can predict a similarity score in one shot and inform the client via an API call and relevant notification through the mobile phone, as depicted in Figure 11.

Figures 12 and 13 showcase the maintenance dashboard that integrate various streams of data, including video and audio feeds, to monitor and report incidents in real-time. The event



FIGURE 11 Facial identification using the ‘follow my kid’ module: (a) Passengers in HOLO vehicle—bounding boxes indicate the ROIs that are passed as input to the network. (b) Passengers in TPG vehicle wearing face masks. Facial characteristics are blurred to preserve the privacy of the passengers.

history log records and timestamps occurrences, such as the detected incidents. These entries provide a sequential overview of the safety-related events detected by the system. Passenger presence is automatically assessed, with onboard passenger count and door status being monitored, the former indicated as ‘2’ suggesting two passengers are onboard, and the latter as ‘Open,’ indicating the vehicle is stationary, in the process of boarding or disembarking passengers. Environmental parameters are also monitored, including air quality index (AQI), temperature, humidity, CO₂, VOC (volatile organic compounds), and particulate matter (PM₂₅) concentration. These metrics suggest a holistic approach to passenger comfort and health, with the current temperature being a comfortable 25.86°C, and the AQI, CO₂, and VOC levels being within a range that is likely considered safe. In a lower section, facial images from the follow my kid database are depicted for use in facial recognition.

6.4 | Comparison with existing in-cabin monitoring solutions

The comparative analysis of three in-cabin monitoring system papers, with a particular emphasis on our solution underscores

its superior approach and methodology in enhancing safety and security within AVs as presented in Table 7. The proposed approach integrates visual and audio analyses and stands out for its high accuracy of 95.2% in abnormal event detection. This methodology demonstrates a comprehensive understanding of the complexities involved in in-cabin monitoring, effectively leveraging multiple data sources to improve the detection and classification of events.

Unlike already existing studies, which either focus narrowly on privacy concerns without extensive performance metrics or present a broad spectrum of functionalities with significant optimization needs for real-time processing, our method offers a balanced and effective solution. More specifically, the approach of Mishra et al. [1], while addressing the crucial issue of privacy preservation through a single AI camera system, falls short by not providing detailed quantitative metrics to assess its effectiveness comprehensively. This lack of detailed performance evaluation could potentially limit its applicability in scenarios requiring high accuracy and reliability. Similarly, their latest system iteration [3], despite offering a comprehensive monitoring framework, needs further optimization for real-time processing, indicating a possible trade-off between the real-time monitoring capabilities and the system’s operational efficiency and responsiveness, which could be detrimental in critical safety applications. In contrast, our proposed system adeptly balances the integration of multiple sensing modalities, achieving not only high accuracy but also detecting different in-cabin events.

7 | CONCLUSIONS

AVs have emerged in modern transportation, driven by advanced deep learning and computer vision technologies. As we approach a future where AVs become common, ensuring passenger safety and comfort within the vehicle’s cabin is of most importance. Our research has introduced a comprehensive in-cabin monitoring framework designed to meet these evolving challenges. By removing human safety officers and leveraging a unified technological foundation, we have presented multiple services aimed at enhancing security, tracking passenger presence, accommodating diverse needs, and personalizing the travel experience. By explicitly integrating augmented safety features with real-time visual and audio analytics, personalized services and real-time updates to reassure vulnerable users, as well as proactive transparency and accountability through services like passenger counting and passenger facial identification, our approach addresses the critical aspects of AV deployment, from safety and security to passenger satisfaction.

While the proposed framework provides real-time, multimodal monitoring with computational efficiency on edge devices, its reliance on pre-defined scenarios and limited scalability for highly complex environments are notable limitations. Additionally, while services like ‘follow my kid/grandparents’ enhance user confidence inside the vehicle, their current scope is limited to onboard monitoring only. Privacy concerns regarding facial capture also present challenges that

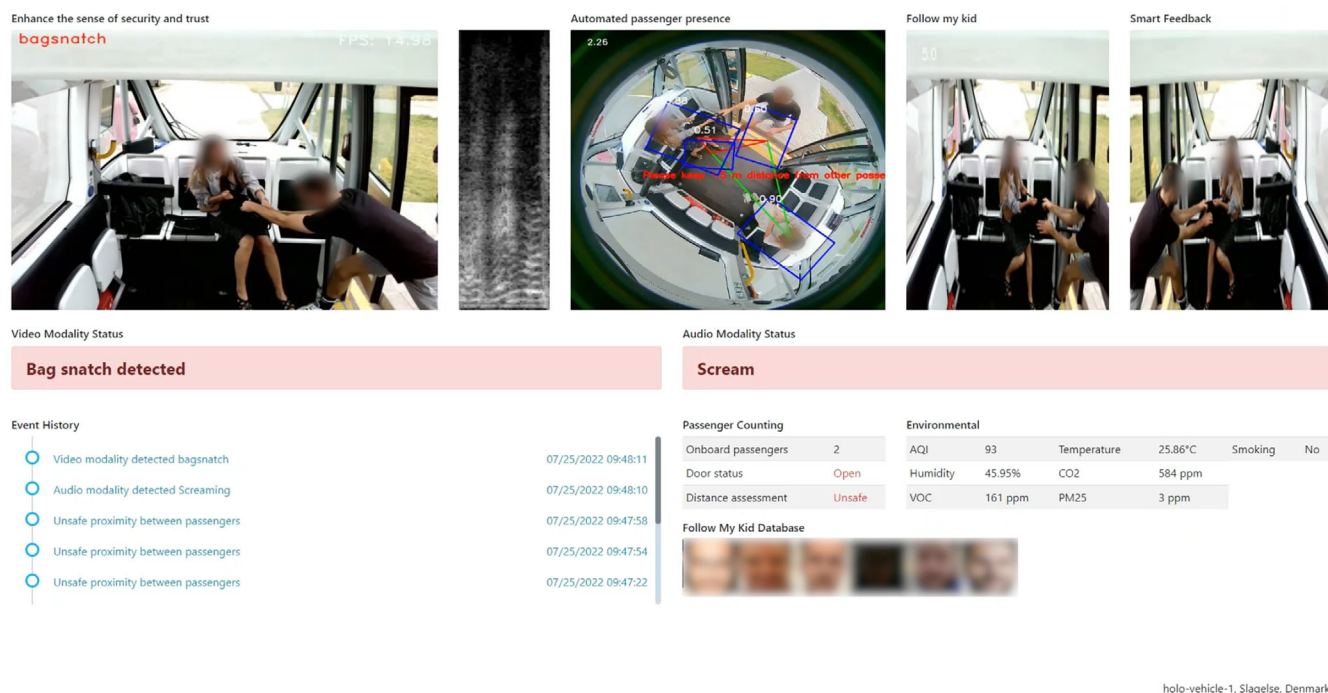


FIGURE 12 Screenshot of the maintenance dashboard showcasing a bag-snatching (video) and screaming (audio) incident at the same time.

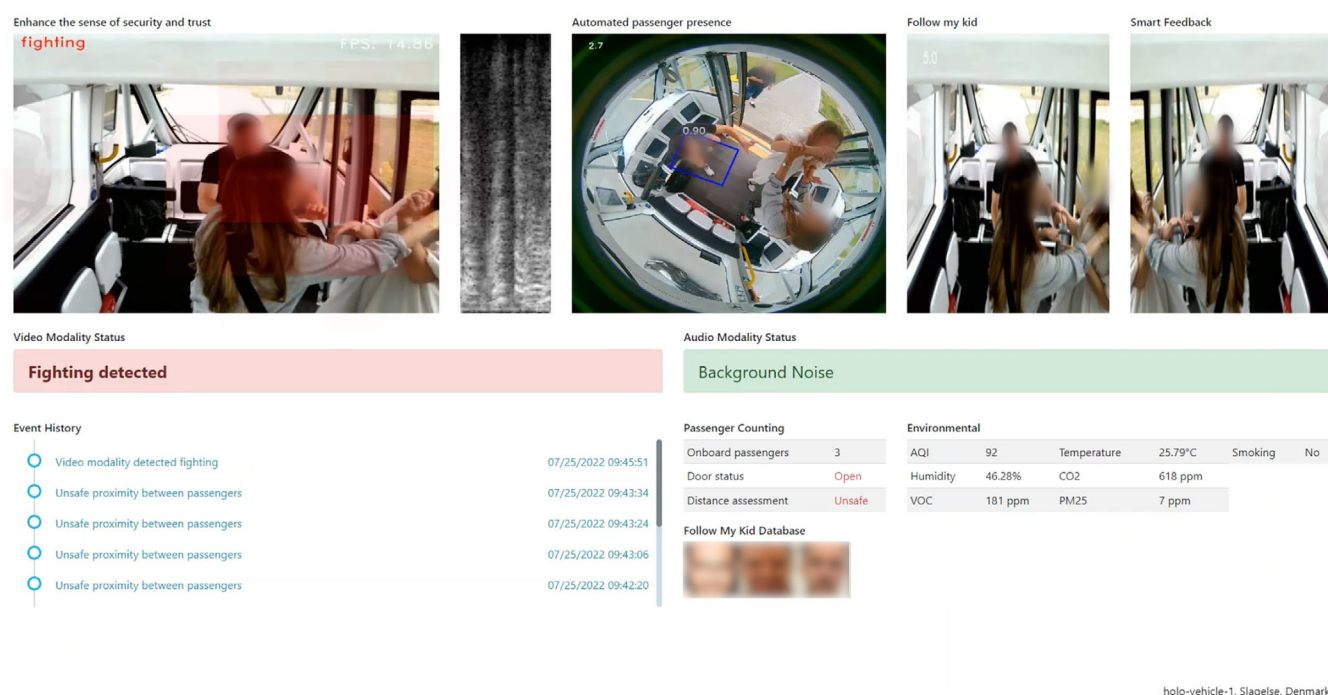


FIGURE 13 Screenshot of the maintenance dashboard showcasing a fighting incident.

will require for additional anonymization techniques in future industrial applications, though our current implementation was specifically designed for research demonstration purposes. As we are transitioning in a future with AVs into the pub-

lic transportation, this framework paves the way for safer, more efficient, and more passenger-centric autonomous transportation systems, increasing user confidence and accelerating public acceptance of AVs. Future research will focus on

TABLE 7 Comparison of the proposed multimodal in-cabin monitoring framework to other similar methods.

Feature	Proposed system	Mishra et al. [1]	Mishra et al. [3]
Framework	Multimodal approach integrating visual and audio analyses	Single AI camera system focusing on privacy-preserved monitoring	Multimodal framework employing advanced sensor technologies
Datasets	CERTH, TPG+HOLO	Custom dataset for AI camera development	Public datasets and custom in-cabin scenarios
Approach	Dual-stream event detection angle-aware passenger detection Siamese Face ID	DenseNet for passenger and object recognition	Various, including DenseNet-121 for audio-based detection
Key results	95.2% accuracy in abnormal event detection	N/A accuracy, emphasizes privacy and efficient monitoring with AI	N/A accuracy in detecting abnormal events and identifying passengers, 96% in object detection
Novelty	Dual-pathway approach for robust event detection	Single-shot monitoring approach with privacy focus	Comprehensive service suite for in-cabin monitoring

benchmarking the framework against state-of-the-art models, improving adaptability to dynamic in-cabin conditions, exploring scalable architectures to support deployment across larger AV fleets, and developing enhanced privacy protection measures to address public concerns while maintaining system functionality.

AUTHOR CONTRIBUTIONS

Dimitris Tsiktsiris: Data curation; investigation; methodology; software; validation; visualization; writing—original draft; writing—review & editing. **Antonios Lalas:** Conceptualization; formal analysis; investigation; methodology; project administration; resources; supervision; validation; visualization; writing—original draft; writing—review & editing. **Minas Dasygenis:** Data curation; resources; software; supervision; validation; visualization; writing—review & editing. **Konstantinos Votis:** Conceptualization; data curation; formal analysis; funding acquisition; investigation; project administration; resources; supervision; writing—review & editing.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Due to privacy considerations, the dataset containing information and images depicting passengers cannot be publicly shared. All data collection was conducted with explicit consent from the individuals involved, ensuring compliance with relevant privacy regulations.

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